ClipSAT : ACT International Subject Test: Mathematics Level 2 [ANSWER KEY]

Unit 1.01: Complex Numbers & De Moivre Theorem

Q1. For the complex number i = sqrt{-1}, which of the following is equivalent to the expression 5 - 3i - 2 - 7i?

[x] A: 3 + 4i

[ ] B: 3 - 10i

[ ] C: 7 + 4i

[ ] D: 3 - 4i

Solution: To subtract complex numbers, distribute the negative sign across the second term and combine like terms real parts with real parts, and imaginary parts with imaginary parts: 5 - 3i - 2 - 7i = 5 - 3i - 2 + 7i Group the real and imaginary components: - Real part: 5 - 2 = 3 - Imaginary part: -3i + 7i = 4i Combining these gives 3 + 4i. Common misconception checks: - Choice B 3 - 10i occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating -3i - 7i = -10i. - Choice C 7 + 4i results from accidentally adding the real parts 5 + 2 = 7 while subtracting the imaginary parts. - Choice D 3 - 4i results from a sign error when computing -3 - -7.

Q2. For what real value of k is the complex expression frac{k + 8i}{3 - i} + frac{2 - ki}{3 + i} a real number?

[x] A: 11

[ ] B: 1

[ ] C: 6

[ ] D: -11

Solution: To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates. The common denominator of the two fractions is: 3 - i3 + i = 3^2 - i^2 = 9 - -1 = 10 Rewrite each term over this denominator: 1. For the first term: frac{k + 8i3 + i}{3 - i3 + i} = frac{3k + ki + 24i + 8i^2}{10} = frac{3k - 8 + k + 24i}{10} 2. For the second term: frac{2 - ki3 - i}{3 + i3 - i} = frac{6 - 2i - 3ki + ki^2}{10} = frac{6 - k + -3k - 2i}{10} Now, sum the real and imaginary components: text{Real part} = frac{3k - 8 + 6 - k}{10} = frac{2k - 2}{10} text{Imaginary part} = frac{k + 24 + -3k - 2}{10} = frac{-2k + 22}{10} For the resulting complex number to be a real number, its imaginary part must equal 0 : frac{-2k + 22}{10} = 0 implies -2k + 22 = 0 implies 2k = 22 implies k = 11 *Common Misconceptions:* - Choosing **1** results from setting the real part equal to 0 2k - 2 = 0 , which would make the expression purely imaginary rather than real. - Choosing **6** results from equating the real and imaginary parts 2k - 2 = -2k + 22 . - Choosing **-11** results from an algebraic sign error when solving -2k = -22 .

Q3. Let z be a complex number satisfying the equation |z - 12| + |z + 12| = 26. What is the maximum possible value of |z - 5i| ?

[x] A: frac{169}{12}

[ ] B: sqrt{194}

[ ] C: 18

[ ] D: 10

Solution: 1. **Identify the geometric locus:** The equation |z - 12| + |z + 12| = 26 represents an ellipse in the complex plane with foci at z_1 = 12 and z_2 = -12 along the real axis. - The focal distance is 2c = 24 implies c = 12 . - The major axis length is 2a = 26 implies a = 13 . - The semi-minor axis length is b = sqrt{a^2 - c^2} = sqrt{13^2 - 12^2} = sqrt{25} = 5 . In Cartesian coordinates z = x + iy , the equation of this ellipse is: frac{x^2}{13^2} + frac{y^2}{5^2} = 1, quad text{with } -13 le x le 13 text{ and } -5 le y le 5. 2. **Express the distance function:** We want to maximize the distance from z to the point 5i = 0, 5 : D^2 = |z - 5i|^2 = x^2 + y - 5^2. From the ellipse equation, x^2 = 169 left1 - frac{y^2}{25}right = 169 - frac{169}{25}y^2 . Substituting x^2 into D^2 : D^2y = 169 - frac{169}{25}y^2 + y^2 - 10y + 25 = -frac{144}{25}y^2 - 10y + 194. 3. **Optimize the quadratic over [-5, 5] :** Since the coefficient of y^2 is negative, D^2y is a downward-opening parabola. The vertex occurs at: y = -frac{-10}{2left-frac{144}{25}right} = -frac{10 times 25}{288} = -frac{125}{144}. Because -5 le -frac{125}{144} le 5 , this critical point lies within the domain. 4. **Calculate the maximum value:** D^2left-frac{125}{144}right = -frac{144}{25}left-frac{125}{144}right^2 - 10left-frac{125}{144}right + 194 = -frac{625}{144} + frac{1250}{144} + 194 = frac{625}{144} + 194 = frac{625 + 27936}{144} = frac{28561}{144}. Taking the square root: D = sqrt{frac{28561}{144}} = frac{sqrt{13^4}}{12} = frac{169}{12}. **Common Misconceptions:** - **Choice B sqrt{194} :** Evaluates the distance at the major

Unit 1.01: Complex Numbers & De Moivre Theorem

Q4. For the imaginary unit i such that i^2 = -1 , what is the product 4 - 3i2 + 5i ?

[x] A: 23 + 14i

[ ] B: -7 + 14i

[ ] C: 23 - 14i

[ ] D: 8 - 15i

Solution: To find the product of the two complex numbers, expand using the distributive property FOIL: 4 - 3i2 + 5i = 42 + 45i - 3i2 - 3i5i Multiply the individual terms: = 8 + 20i - 6i - 15i^2 Substitute i^2 = -1 : = 8 + 14i - 15-1 = 8 + 14i + 15 = 23 + 14i Common errors: - -7 + 14i results from forgetting that i^2 = -1 changes the sign of -15i^2 , incorrectly computing 8 - 15 = -7 . - 8 - 15i results from only multiplying the first terms and last terms, omitting the cross terms.

Q5. For the imaginary unit i, where i^2 = -1, the expression frac{k + 6i}{3 - 2i} is equal to a real number for some real constant k. What is the value of k?

[x] A: -9

[ ] B: -4

[ ] C: 4

[ ] D: 9

Solution: To simplify the quotient of two complex numbers, multiply the numerator and denominator by the complex conjugate of the denominator, 3 + 2i: frac{k + 6i}{3 - 2i} = frac{k + 6i3 + 2i}{3 - 2i3 + 2i} Expand the denominator: 3 - 2i3 + 2i = 3^2 - 2i^2 = 9 - 4-1 = 9 + 4 = 13 Expand the numerator: k + 6i3 + 2i = 3k + 2ki + 18i + 12i^2 = 3k + 12-1 + 2k + 18i = 3k - 12 + 2k + 18i Thus, the expression can be written in standard form a + bi as: frac{3k - 12}{13} + frac{2k + 18}{13}i For this expression to be a real number, its imaginary part must be zero: frac{2k + 18}{13} = 0 implies 2k + 18 = 0 implies 2k = -18 implies k = -9 Common Misconceptions: - Setting the real part to zero confusing "real number" with "purely imaginary" gives 3k - 12 = 0 implies k = 4. - Making a sign error when multiplying by the conjugate leads to -2k + 18 = 0 implies k = 9 or 3k + 12 = 0 implies k = -4.

Q6. In the complex plane, the complex numbers z and w satisfy |z| = 5, |w| = 7, and |z + w| = 10. What is the value of |z - w|?

[x] A: 4sqrt{3}

[ ] B: 2sqrt{6}

[ ] C: 4sqrt{6}

[ ] D: 2sqrt{37}

Solution: To find |z - w|, we use the algebraic properties of the modulus and complex conjugates, which correspond to the parallelogram law in the complex plane. Recall that for any complex number u, |u|^2 = uoverline{u}. Expanding the modulus squared of the sum and difference of z and w: 1. |z + w|^2 = z + woverline{z} + overline{w} = |z|^2 + |w|^2 + zoverline{w} + overline{z}w 2. |z - w|^2 = z - woverline{z} - overline{w} = |z|^2 + |w|^2 - zoverline{w} + overline{z}w Adding these two equations eliminates the cross terms zoverline{w} + overline{z}w: |z + w|^2 + |z - w|^2 = 2|z|^2 + |w|^2 Substitute the given values |z| = 5, |w| = 7, and |z + w| = 10: 10^2 + |z - w|^2 = 2left5^2 + 7^2right 100 + |z - w|^2 = 225 + 49 = 274 = 148 |z - w|^2 = 148 - 100 = 48 Taking the principal square root gives: |z - w| = sqrt{48} = 4sqrt{3} Common Misconceptions: - Choice B 2sqrt{6} results from incorrectly computing sqrt{|w|^2 - |z|^2} = sqrt{49 - 25} = sqrt{24}. - Choice C 4sqrt{6} arises from mistakenly computing sqrt{10^2 - 7 - 5^2} = sqrt{96}. - Choice D 2sqrt{37} comes from calculating sqrt{2|z|^2 + |w|^2} = sqrt{148} while forgetting to subtract |z + w|^2.

Unit 1.01: Complex Numbers & De Moivre Theorem

Q7. For the imaginary unit i, where i^2 = -1, which of the following complex numbers is equivalent to 4 + 3i2 - 5i?

[x] A: 23 - 14i

[ ] B: 23 + 14i

[ ] C: -7 - 14i

[ ] D: -7 + 14i

Solution: To multiply the two complex numbers, apply the distributive property FOIL: 4 + 3i2 - 5i = 42 + 4-5i + 3i2 + 3i-5i Compute each term: = 8 - 20i + 6i - 15i^2 Since i^2 = -1, substitute -15i^2 = -15-1 = 15: = 8 - 14i + 15 = 8 + 15 - 14i = 23 - 14i A common error is setting i^2 = 1, which incorrectly yields 8 - 15 - 14i = -7 - 14i.

Q8. In the complex number system, let i^2 = -1. For what real value of k is the expression frac{6 + ki}{3 + 2i} a purely imaginary number?

[x] A: -9

[ ] B: -4

[ ] C: 4

[ ] D: 9

Solution: To determine when the quotient is purely imaginary, multiply the numerator and denominator by the complex conjugate of the denominator, 3 - 2i: frac{6 + ki}{3 + 2i} = frac{6 + ki3 - 2i}{3 + 2i3 - 2i} Expand the denominator using i^2 = -1: 3 + 2i3 - 2i = 3^2 - 2i^2 = 9 - 4-1 = 13 Expand the numerator: 6 + ki3 - 2i = 18 - 12i + 3ki - 2ki^2 = 18 + 2k + 3k - 12i Thus, the expression in standard form a + bi is: frac{18 + 2k}{13} + frac{3k - 12}{13}i A complex number is purely imaginary if and only if its real part is equal to 0 and its imaginary part is nonzero: frac{18 + 2k}{13} = 0 implies 18 + 2k = 0 implies k = -9 Checking the imaginary part for k = -9: frac{3-9 - 12}{13} = frac{-39}{13} = -3 neq 0 So the expression simplifies to -3i, which is purely imaginary. Common error: Setting the imaginary part to 0 3k - 12 = 0 implies k = 4 makes the number purely real, not purely imaginary.